JOHN JOSEPH YOUKHANA

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EDUCATION

University of Illinois at Urbana-Champaign GPA 3.1/4.0
Bachelor of Science in Electrical Engineering Expected Graduation: May 2027
Relevant Coursework

(ECE 486) Control Systems	(ECE 483) Analog IC Design	(ECE 330) Power Circuits & Electromechanics
(ECE 340) Semiconductor Devices	(ECE 342) Electronic Circuits	(ECE 385) Digital Systems Laboratory
(ECE 329) Fields and Waves I	(ECE 310) Digital Signal Processing	(CS 225) Data Structures & Algorithms

Wilbur Wright College GPA 3.74/4.0
Associate of Engineering Science Aug 2022 - May 2024

PROJECTS

Reaction Wheel Pendulum Control Systems Project Completed May 2026

- Designed a real-time control system for a reaction wheel pendulum to achieve swing-up stabilization of a pendulum arm using the reaction torque generated by a DC motor driven flywheel.
- Derived nonlinear system model, conducted system identification to estimate plant parameters through experimental data analysis.
- Implemented real-time friction compensation in Simulink using PI velocity control to estimate and compensate coulomb and viscous friction effects, to achieve frictionless motor performance.
- Designed and implemented three-state feedback control to stabilize pendulum in the inverted position using reaction wheel torque.
- Designed and tested an observer-based controller to estimate unmeasured velocity states and improve real-time stabilization performance.
- Achieved swing-up, inverted stabilization, and switching control between downward and inverted equilibria on embedded hardware (TI-C2000 + Simulink) using friction compensation, state feedback, and observer-based state estimation.

Low-Dropout Regulator Analog Design Project Expected May 2026

- Designing CMOS LDO in Cadence in 180nm process technology. The device will supply sensitive analog devices with 1.0-1.4V for 0.1-10mA of load current.
- Implemented LDO error amplifier using folded-cascode amplifier architecture with biasing circuit. The amplifier is designed to satisfy phase margin, output capacitance, and DC load/line regulation requirements.

AC-DC Power Supply and Voltage Regulator Fall 2025

- Developed a multi-stage AC-DC converter consisting of transformation, full bridge rectification, filter stage, regulation, and amplification stages
- Designed to step down 120VAC to 25VAC. Converts 25VAC to a stable ~9.5 DC voltage, with low noise and ripple.

RELEVANT EXPERIENCE

Illini Solar Car Champaign, IL
Electrical Team member Aug 2024 - Present

- Designed and tested custom PCBs in KiCad for high/low voltage systems, conducting rigorous testing to verify functionality and board integrity.
- Contributed to design of a battery charge switch PCB using KiCad, incorporating digital logic for high-voltage switching in the team's race car.

SKILLS

Skills: PCB Layout & Assembly, Control System Design, Power systems, Electronics Design, Amplifier Design, Digital design, Signal Processing, FPGA Programming, Verification/Testing
Tools: MATLAB, Simulink, Cadence Virtuoso, LTspice, MS Excel, Xilinx Vivado, Fusion360, KiCad
Programming: Python, C/C++, SystemVerilog, VHDL, LabVIEW, Assembly
Interests: Microcontroller projects, Competitive Programming, Quantitative Finance, Golf, Rock Climbing

LICENSES

Amateur Radio Technician Class Operator Effective date: April 2026
Federal Communications Commission Callsign: KE9FKZ